



VECTOR SPACES

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CLASS 12 : CONTENT



➤ Problems on Uniqueness, Existence of right and left inverse

Matrix of rank 1

RIGHT INVERSE

Problem 1: Find the left or Right Inverse for the following matrices, whichever exists:-

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

Solution:- step I: Apply Gauss elimination and obtain rank

Step II: check if rank 'r' of the matrix is equal to 'm' or 'n' i.e. number of rows or columns of the matrix A.

step III: if $r(A) = m$ then (Right Inverse exists) $A_{m \times n} B_{n \times m} = I_{m \times m}$

if $r(A) = n$ then (Left inverse exists) $C_{n \times m} A_{m \times n} = I_{n \times n}$

RIGHT INVERSE

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

$\rho(A) = 2 = m \quad \therefore \text{Right inverse exists}$

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

Best Right Inverse is $A^T (A A^T)^{-1}$

$$A A^T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}; \quad (A A^T)^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

LEFT INVERSE

$$A^T(AA^T)^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 2/3 & -1/3 \\ 1/3 & 1/3 \\ -1/3 & 2/3 \end{pmatrix} \text{Ans}$$

2) Find inverse for the matrix $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}$

Solution: $\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$

$\rho(A) = 2 = n = \text{number of columns}$

A has left inverse; Best left inverse is $(A^T A)^{-1} A^T$

$$(A^T A)^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}; (A^T A)^{-1} A^T = \begin{pmatrix} 2/3 & 1/3 & -1/3 \\ -1/3 & 1/3 & 2/3 \end{pmatrix} \text{Ans}$$

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MATRIX OF RANK 1



Matrices Of Rank One:

When the rank of a matrix is as small as possible,
a complicated system of equations can be broken into simple pieces.
Those simple pieces are matrices of rank one.

The matrix has rank $r = 1$.

We can write such matrices as a column times row.

That is

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & 2 & 2 \\ 6 & 3 & 3 \\ 8 & 4 & 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \end{bmatrix} \Rightarrow A = UV^T$$

$u = (1, 2, 3, 4)$ $v = (2, 1, 1)$



THANK YOU

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